Complete proofs
Infinite NPA depth at minimal semidefinite complexity

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Alice and Bob each choose one of two measurements and obtain an outcome in $\{\pm 1\}$. Write A_0, A_1 and B_0, B_1 for their observables, and $\mathcal{Q}_{222}$ for the set of quantum correlations in this scenario. Let $M = \langle A_0 \rangle + \langle B_0 \rangle$ and $S = \langle A_0B_0 + A_0B_1 + A_1B_0 - A_1B_1 \rangle$ in the two-input, two-output Bell scenario, and let $\mathcal{W} = \{(M, S) : p \in \mathcal{Q}_{222}\}$. A semidefinite lift describes membership by introducing auxiliary variables and imposing affine equations and positive semidefinite matrix constraints. Its block orders are the sizes of these matrices. The semidefinite extension degree is the smallest possible largest block order, allowing any finite number of blocks. We give an explicit lift of $\mathcal{W}$ with blocks of orders five and four and prove that its extension degree is two. Every fixed level of the Navascués–Pironio–Acín (NPA) hierarchy nevertheless projects to a strict superset of $\mathcal{W}$.

For each supporting inequality, the quantum maximum can be attained when Alice's two observables and Bob's two observables have the same relative angle. Part I proves this reduction and constructs the membership matrices from pure two-qubit states. It also proves that two such strategies suffice, treats arbitrary signs and unequal marginal coefficients, and gives thirty-two equivalent projections under measurement relabellings. Part II proves that no fixed NPA level is exact. If its error remained bounded relative to the quantum violation near the local boundary, the certificate would give an impossible polynomial sum of squares.

I. The finite representation

F1 Why equal local measurement angles suffice

The supporting inequalities of $\mathcal{W}$ have the form $aM + hS \leq \lambda$. Write $\mathrm{CHSH} = A_0B_0 + A_0B_1 + A_1B_0 - A_1B_1$ for the CHSH operator. We determine when its tilted form is bounded above by λI for every choice of state and measurements. For $H = h\,\mathrm{CHSH} + a(A_0 + B_0)$, with $h, a \in \mathbb{R}$, write

$$\mathcal{C} = \{(\lambda, a, h) : \lambda I - H \succeq 0 \text{ for every quantum realization}\}.$$

Projective dilation and Jordan's lemma reduce this condition to real two-qubit blocks. With X, Y, Z denoting the Pauli matrices, put

$$A_0 = B_0 = Z, \quad A_1 = cZ + rX, \quad B_1 = dZ + sX, \qquad r^2 = 1 - c^2,\ s^2 = 1 - d^2.$$

Here $c, d \in [-1, 1]$ are the cosines of the two local measurement angles. We may take $r, s \geq 0$ by local Z conjugations. Endpoints include the one-dimensional blocks. Complex state vectors cause no further condition, because the Bell matrix is real symmetric.

Set $H_* = |h|$, $b = |a|$ and $L = 2H_* + 2b$. Deterministic assignments attain L: choose $A_0 = B_0 = \mathrm{sign}(a)$, and take $A_1 = B_1$ equal to that sign for $h \geq 0$, or its negative for $h < 0$. At $a = 0$, either sign works. Therefore every admissible λ satisfies $\lambda \geq L$.

The determinant controls positivity. With $\Omega = 1 + c + d - cd \in [-2, 2]$, the coefficients in

$$\det(uI + \lambda I - H) = u^4 + e_1 u^3 + e_2 u^2 + e_3 u + e_4$$

are

$$\begin{aligned} e_1 &= 4\lambda, & e_2 &= 6\lambda^2 - 4a^2 - 8h^2, \\ e_3 &= 4\lambda^3 - 8\lambda a^2 - 16\lambda h^2 - 8ha^2\Omega, & e_4 &= \det(\lambda I - H). \end{aligned}$$

The first three are nonnegative for $\lambda \geq L$. For e_3, bound its last term below by $-16H_* b^2$. The resulting cubic is increasing for $\lambda \geq L$, and at L equals $16b(b + 2H_*)^2$. Consequently

$$\lambda I - H \succeq 0 \quad \Longleftrightarrow \quad e_4 \geq 0.$$

Indeed, a negative eigenvalue would produce a positive root of a polynomial whose coefficients are all nonnegative. This argument includes zero determinants and multiple zero eigenvalues.

At fixed product, minimize the sum. Put $x = (1 - c)/2$, $y = (1 - d)/2$, so $(x, y) \in [0, 1]^2$, and denote the determinant $\det(\lambda I - H)$ by $D(x, y)$. Direct expansion gives

$$\begin{aligned} D(x, y) &= D_0 + 64h^2(4h^2 - a^2)xy(x + y) \\ &\quad + (32\lambda ha^2 + 128h^2a^2 - 256h^4)xy - 256h^4x^2y^2, \\ D_0 &= [(\lambda - 2h)^2 - 4a^2](\lambda + 2h)^2. \end{aligned}$$

Only $x + y$ varies when xy is fixed. Arithmetic–geometric mean gives $x + y \geq 2\sqrt{xy}$, and therefore

$$D(x, y) - D(\sqrt{xy}, \sqrt{xy}) = 64h^2(4h^2 - a^2)xy(\sqrt{x} - \sqrt{y})^2. \tag{F1}$$

For $|a| \leq |h|$, the right side is nonnegative. Replacing (x, y) by $(\sqrt{xy}, \sqrt{xy})$ stays in $[0, 1]^2$ and makes $c = d$. Thus nonnegative determinants for all equal-angle choices imply nonnegative determinants for every pair c, d, including the endpoints. For $\lambda \geq L$, this is equivalent to positivity of the corresponding slacks.

The remaining directions are classical. The range $b \geq H_*$ has exact support value L. To prove this, set $C_B = (B_0 + B_1)/2$, $D_B = (B_0 - B_1)/2$. Then

$$4I - (\mathrm{CHSH} + 2A_0) = (I - C_B - A_1 D_B)^2 + \frac{1}{4}\sum_{y=0}^{1}[(I - A_0)(I + B_y)]^2 \succeq 0. \tag{F2}$$

Here $C_B^2 + D_B^2 = I$ and $C_B D_B + D_B C_B = 0$. Party exchange and output flips give $\sigma\mathrm{CHSH} + 2\tau A_0 \leq 4I$ and $\sigma\mathrm{CHSH} + 2\tau B_0 \leq 4I$, for either sign σ, τ. Average these inequalities, multiply by H_*, and add $(b - H_*)[2I - \tau(A_0 + B_0)] \succeq 0$. The resulting bound is $H \leq LI$, attained by the deterministic strategies above. This also covers $h = 0$, the critical line, and both coefficient signs.

The quartic matrix. On $c = d$, the Bell basis

$$\phi_\pm = (|00\rangle \pm |11\rangle)/\sqrt{2}, \qquad \psi_\pm = (|01\rangle \pm |10\rangle)/\sqrt{2}$$

splits off ψ_-, with eigenvalue $-2hc$. In the basis (ϕ_+, ϕ_-, ψ_+), the remaining block is

$$H_3(c) = \begin{pmatrix} 2hc & 2a & 0 \\ 2a & 2h(1 + c - c^2) & 2h(1 - c)\sqrt{1 - c^2} \\ 0 & 2h(1 - c)\sqrt{1 - c^2} & -2h(1 + c - c^2) \end{pmatrix}.$$

To remove the square root, use $c = (1-t^2)/(1+t^2)$ and multiply the slack $\lambda I_3 - H_3(c)$ by $(1+t^2)^2$. Writing

$$p_0 = (1+t^2)^2, \qquad p_1 = 1-t^4, \qquad p_2 = 1+4t^2-t^4,$$

the resulting quartic is

$$K_{h,a,\lambda}(t) = \begin{pmatrix} \lambda p_0 - 2hp_1 & -2ap_0 & 0 \\ -2ap_0 & \lambda p_0 - 2hp_2 & -8ht^3 \\ 0 & -8ht^3 & \lambda p_0 + 2hp_2 \end{pmatrix}. \tag{F3}$$

Finite $t \geq 0$ covers $-1 < c \leq 1$; $c = -1$ follows by continuity. Negative t adds no condition because $K(-t) = \operatorname{diag}(1,1,-1)K(t)\operatorname{diag}(1,1,-1)$.

Conversely, $K(t) \succeq 0$ for all real t forces the local bound. Its upper-left block at $t = 0$ gives $\lambda \geq 2h + 2|a|$; the leading coefficient at $t = \infty$ gives $\lambda \geq -2h + 2|a|$. Thus $\lambda \geq L$, which also bounds the omitted eigenvalue $-2hc$. Equation (F1) applies for $|a| \leq |h|$, and Eq. (F2) gives the bound for $|a| \geq |h|$. We have proved, for all real coefficients,

$$(\lambda, a, h) \in \mathcal{C} \quad \Longleftrightarrow \quad K_{h,a,\lambda}(t) \succeq 0 \quad \text{for every } t \in \mathbb{R}. \tag{F4}$$

A fixed affine Gram lift. Every real univariate matrix polynomial that is positive semidefinite (PSD) for all real arguments can be written as a sum $\sum_j F_j(t)^T F_j(t)$ of polynomial matrix squares, including identically singular ones [24]. Since the diagonal entries of K have degree at most four, every square factor has degree at most two: highest-degree real squares cannot cancel. For $\xi = (\xi_1, \xi_2, \xi_3)^T \in \mathbb{R}^3$, a Gram representation expresses $\xi^T K(t)\xi$ as a quadratic form in monomials in t and ξ. Average this representation under $(t, \xi) \mapsto (-t, \operatorname{diag}(1,1,-1)\xi)$. The even and odd monomials then form separate blocks, with vectors

$$Z_+ = (\xi_1, \xi_2, t\xi_3, t^2\xi_1, t^2\xi_2)^T, \qquad Z_- = (\xi_3, t\xi_1, t\xi_2, t^2\xi_3)^T.$$

Thus $\xi^T K(t)\xi = Z_+^T G Z_+ + Z_-^T J Z_-$, with $G \succeq 0$ of order five and $J \succeq 0$ of order four. The sixteen coefficient equations are

$$\begin{aligned} g_{11} &= \lambda - 2h, & g_{12} &= -2a, \\ g_{22} &= \lambda - 2h, & j_{11} &= \lambda + 2h, \\ g_{13} + j_{12} &= 0, & g_{23} + j_{13} &= 0, \\ 2g_{14} + j_{22} &= 2\lambda, & g_{15} + g_{24} + j_{23} &= -4a, \\ 2g_{25} + j_{33} &= 2\lambda - 8h, & g_{33} + 2j_{14} &= 2\lambda + 8h, \\ g_{34} + j_{24} &= 0, & g_{35} + j_{34} &= -8h, \\ g_{44} &= \lambda + 2h, & g_{45} &= -2a, \\ g_{55} &= \lambda + 2h, & j_{44} &= \lambda - 2h. \end{aligned} \tag{F5}$$

These equations include zero coefficients, lower polynomial degrees and every rank-deficient case. Setting $h = 1$ describes all pairs (a, λ) for which $S + aM \leq \lambda$ holds on $\mathcal{W}$. Restricting to $0 \leq a \leq 1$ requires the additional scalar inequalities $a \geq 0$ and $a \leq 1$.

F2 Explicit membership matrices for the quantum region

Introduce sixteen real variables $z_1, \ldots, z_{16}$ and define

$$U(z) = \begin{pmatrix} z_1 & z_2/2 & z_5/2 & z_7 & z_8/2 \\ z_2/2 & z_3 & z_6/2 & z_8/2 & z_9 \\ z_5/2 & z_6/2 & z_{10} & z_{11}/2 & z_{12}/2 \\ z_7 & z_8/2 & z_{11}/2 & z_{13} & z_{14}/2 \\ z_8/2 & z_9 & z_{12}/2 & z_{14}/2 & z_{15} \end{pmatrix},$$
$$V(z) = \begin{pmatrix} z_4 & z_5/2 & z_6/2 & z_{10} \\ z_5/2 & z_7 & z_8/2 & z_{11}/2 \\ z_6/2 & z_8/2 & z_9 & z_{12}/2 \\ z_{10} & z_{11}/2 & z_{12}/2 & z_{16} \end{pmatrix}. \tag{F6}$$

Membership is equivalent to feasibility of the following semidefinite program (SDP):

$$(M, S) \in \mathcal{W} \quad \Longleftrightarrow \quad \text{there exists } z \in \mathbb{R}^{16} \text{ with } U(z) \succeq 0, \quad V(z) \succeq 0,$$

and

$$\begin{aligned} 1 &= z_1 + z_3 + z_4 + 2z_7 + 2z_9 + 2z_{10} + z_{13} + z_{15} + z_{16}, \\ M &= 2z_2 + 4z_8 + 2z_{14}, \\ S &= 2z_1 + 2z_3 - 2z_4 + 8z_9 - 8z_{10} + 8z_{12} - 2z_{13} - 2z_{15} + 2z_{16}. \end{aligned} \tag{F7}$$

We first construct feasible z from two-qubit states, proving that every point in $\mathcal{W}$ passes this test. Use $u, v \geq 0$, $u^2+v^2 = 1$, and a real unit vector $\xi = (\xi_1, \xi_2, \xi_3)$ in the triplet basis (ϕ_+, ϕ_-, ψ_+). The common local measurement angle has cosine $c = u^2 - v^2$, with sine $2uv$. Define

$$r_+ = (u^2\xi_1, u^2\xi_2, uv\xi_3, v^2\xi_1, v^2\xi_2)^T, \qquad r_- = (u^2\xi_3, uv\xi_1, uv\xi_2, v^2\xi_3)^T.$$

The matrices $U = r_+ r_+^T$, $V = r_- r_-^T$ each have rank at most one and have precisely the shared entries in Eq. (F6). Explicitly, choose

$$\begin{aligned} z = (&u^4\xi_1^2,\ 2u^4\xi_1\xi_2,\ u^4\xi_2^2,\ u^4\xi_3^2, \\ &2u^3v\xi_1\xi_3,\ 2u^3v\xi_2\xi_3,\ u^2v^2\xi_1^2,\ 2u^2v^2\xi_1\xi_2, \\ &u^2v^2\xi_2^2,\ u^2v^2\xi_3^2,\ 2uv^3\xi_1\xi_3,\ 2uv^3\xi_2\xi_3, \\ &v^4\xi_1^2,\ 2v^4\xi_1\xi_2,\ v^4\xi_2^2,\ v^4\xi_3^2). \end{aligned} \tag{F8}$$

Let $\ell(z)$ denote the linear form on the right-hand side of the normalization equation in Eq. (F7). Substitution gives $\ell(z) = 1$ and

$$\begin{aligned} M &= 4\xi_1\xi_2, \\ S &= 2(u^4 - v^4)\xi_1^2 + 2(u^4 + 4u^2v^2 - v^4)(\xi_2^2 - \xi_3^2) + 16uv^3\xi_2\xi_3. \end{aligned}$$

These are the state's observed averages. The variables u, v include both endpoints without division or limits.

Let $\mathcal{T}$ be the set of pairs (M, S) obtained from these real states in the triplet subspace. It is compact and path-connected, being a continuous image of the quarter-circle $u, v \geq 0$, $u^2 + v^2 = 1$, times the unit sphere S^2. The singlet contributes only $(0, -2c)$, and

$$(0, -2c) = \frac{1+c}{2}(0, -2) + \frac{1-c}{2}(0, 2).$$

Both endpoints belong to $\mathcal{T}$: choose $\xi = (0, 0, 1)$ and respectively $(u, v) = (1, 0), (0, 1)$. Since choices with $c = d$ attain every support value, $\operatorname{conv}\mathcal{T}$ and $\mathcal{W}$ have the same supporting inequalities. They are compact convex sets, hence

$$\mathcal{W} = \operatorname{conv}\mathcal{T}. \tag{F9}$$

Convex mixtures of the displayed witnesses therefore admit every point of $\mathcal{W}$ into the SDP, including the boundary.

To prove that every feasible z gives a point in $\mathcal{W}$, write Eq. (F5) as $\mathcal{A}(G, J) = B(\lambda, a, h)$ in the displayed row order. Here $\mathcal{A}$ maps the two Gram matrices to the sixteen left-hand sides, and B maps (λ, a, h) to the corresponding right-hand sides. With the trace inner product on symmetric matrices, the adjoint of $\mathcal{A}$ satisfies $(U(z), V(z)) = \mathcal{A}^* z$ and $B^T z = (\ell(z), -M(z), -S(z))$. Any feasible z satisfies

$$\lambda - aM(z) - hS(z) = \mathrm{tr}[U(z)G] + \mathrm{tr}[V(z)J] \geq 0$$

for every valid supporting inequality, by Eq. (F4) and its Gram representation. Thus its coordinates belong to $\mathcal{W}$. This proves the membership test without invoking dual attainment.

Two strategies suffice. Every point of the convex hull of a compact connected set in the plane lies on a segment joining two points of the set. Here is a direct proof. Take $p \in \mathrm{conv}\,\mathcal{T} \setminus \mathcal{T}$ and project $\mathcal{T}$ radially onto the unit circle about p. Its image is compact and connected. If it contains no antipodal pair, it is an arc of length less than π, contained in an open semicircle. The midpoint direction then strictly separates p from $\mathcal{T}$, a contradiction. Thus $x - p = rn$ and $y - p = -sn$ for some $x, y \in \mathcal{T}$ and $r, s > 0$, and $p = (sx + ry)/(r + s)$.

Applied to Eq. (F9), this shows that every $(M, S) \in \mathcal{W}$ is realized by a mixture of at most two real pure two-qubit strategies, each with equal local measurement angles. Their witnesses give simultaneously $\mathrm{rank}\, U, \mathrm{rank}\, V \leq 2$. If the point is extreme, it belongs to $\mathcal{T}$ and both ranks can be at most one. These rank bounds describe available witnesses; no rank constraint is added to the convex SDP.

Compactness and the dual formulation. The normalized feasible auxiliary set is also compact. The following argument gives an alternative dual proof of membership and can be useful when optimizing over the lift. Define the dual image

$$\mathcal{D} = \{B^T z : U(z) \succeq 0,\ V(z) \succeq 0\}$$

and note that it is closed. Indeed, the positive definite matrices

$$G_0 = \mathrm{diag}(1, 1, 2, 1, 1), \qquad J_0 = \mathrm{diag}(1, 2, 2, 1)$$

satisfy $\mathcal{A}(G_0, J_0) = B(1, 0, 0)$, and hence $\ell(z) = \mathrm{tr}[U(z)G_0] + \mathrm{tr}[V(z)J_0]$. If $B^T z$ converges, this weighted trace bounds both PSD matrices. Every z_i occurs in their entries, so z is bounded and has a convergent subsequence. The limit is feasible. Let $\mathcal{C}^*$ denote the dual cone, whose elements have nonnegative inner product with every element of $\mathcal{C}$. The strictly feasible pair (G_0, J_0) allows us to take the dual of the Gram representation, giving $\mathcal{C}^* = \overline{\mathcal{D}} = \mathcal{D}$. Finally, the compact convex set $\mathcal{W}$ is the intersection of its support half-spaces, so $(M, S) \in \mathcal{W}$ exactly when $(1, -M, -S) \in \mathcal{C}^*$. This proves the membership test. The normalized feasible z-set is itself compact by the same trace argument.

F3 Minimal block order and the NPA comparison

The set $\mathcal{W}$ is compact and convex. The explicit lift also makes it semialgebraic: its membership condition can be expressed by a finite combination of polynomial equalities and inequalities. Writing $\mathrm{sxdeg}(\mathcal{W})$ for its semidefinite extension degree, Scheiderer's planar theorem gives $\mathrm{sxdeg}(\mathcal{W}) \leq 2$ [25]. A short argument shows that $\mathcal{W}$ is not a polygon.

Let $f(a) = \max_{\mathcal{W}}(S + aM)$, and $w = 2\sqrt{2} = f(0)$. The identity

$$\mathrm{CHSH}(c, d)^2 = 4I + 4rs\, Y \otimes Y$$

shows that attaining w requires $r = s = 1$. At $c = d = 0$, the top state is $v = \cos(\pi/8)\phi_- + \sin(\pi/8)\psi_+$, with both reduced states $I/2$. Thus every CHSH-maximizing correlation has $M = 0$. For maximizers p_a of $f(a)$, compactness then gives $M(p_a) \to 0$ as $a \downarrow 0$, and

$$0 \le \frac{f(a) - w}{a} \le M(p_a) \longrightarrow 0.$$

Yet $f(a) > w$ for every $a > 0$: keeping $c = d = 0$, the normalized state $(v + \varepsilon\phi_+)/\sqrt{1+\varepsilon^2}$ gives

$$\frac{w + 4a\cos(\pi/8)\varepsilon}{1+\varepsilon^2} > w \quad \text{for} \quad \varepsilon = \frac{2a\cos(\pi/8)}{w}.$$

A polygon would make f piecewise affine with finitely many pieces. Its zero right derivative at zero would force an initial constant piece, contradicting this strict increase. Therefore $\mathcal{W}$ is nonpolyhedral, and

$$\boxed{\mathrm{sxdeg}(\mathcal{W}) = 2.}$$

The same proof gives extension degree two for $\{(a, \lambda) : 0 \le a \le 1,\ \lambda \ge f(a)\}$.

Write $\omega^{(L)}_{\mathrm{NPA}}(a)$ for the level-L NPA upper bound on $S + aM$. The prior NPA result applies to these support directions $h = 1$ [26], with $T = 1 - a \downarrow 0$:

$$f(1-T) = 4 - 2T + \frac{4}{3}T^3 + O(T^4), \qquad \frac{\omega^{(L)}_{\mathrm{NPA}}(1-T) - f(1-T)}{T^3} \longrightarrow +\infty$$

for every fixed finite level L. Consequently no fixed NPA projection onto (M, S) equals $\mathcal{W}$, while Eqs. (F6)–(F7) describe that same region exactly. The nontermination theorem is proved in Part II. The finite lift and the nonpolyhedrality argument above are independent of that proof.

F4 All signed asymmetric tilt branches

The determinant minimization extends to arbitrary real marginal coefficients in

$$H_{\alpha,\beta} = \mathrm{CHSH} + \alpha A_0 + \beta B_0.$$

Put $q = |\alpha| + |\beta|$. Its local bound is $2 + q$, attained for any tilt signs. On $q = 2$, combine the signed one-sided bounds from Eq. (F2) with weights $|\alpha|/2$ and $|\beta|/2$. This gives the same quantum bound. For $q > 2$, first apply this result to coefficients $2\alpha/q, 2\beta/q$. The remaining marginal terms are bounded above by $q - 2$, giving $2 + q$ again.

Suppose $q < 2$ and $\lambda \ge 2 + q$. On a two-qubit Jordan block, the coefficients of u^2 and u in $\det(uI + \lambda I - H_{\alpha,\beta})$ are

$$e_2 = 6\lambda^2 - 2\alpha^2 - 2\beta^2 - 8, \qquad e_3 = 4\lambda^3 - 4\lambda(\alpha^2 + \beta^2 + 4) - 8\alpha\beta\Omega.$$

Both are nonnegative. For e_3, a lower bound is $g(\lambda) = 4\lambda^3 - 4\lambda(q^2+4) - 4q^2$. It increases on this range, and $g(2+q) = 32q + 12q^2 \ge 0$. Together with $e_1 = 4\lambda \ge 0$, these inequalities show that the determinant again decides positive semidefiniteness.

With $x = (1-c)/2$, $y = (1-d)/2$, direct expansion gives

$$\begin{aligned} D(x,y) &= D_0 + xy[C + 64Ax + 64By - 256xy], \\ A &= 4 - \alpha^2, \qquad B = 4 - \beta^2, \\ C &= 32\lambda\alpha\beta + 64\alpha^2 + 64\beta^2 - 256, \\ D_0 &= [(\lambda-2)^2 - (\alpha+\beta)^2][(\lambda+2)^2 - (\alpha-\beta)^2]. \end{aligned}$$

Both A, B are positive. Exchange parties if necessary so that $|\alpha| \ge |\beta|$, and set $\rho = A/B \in (0, 1]$. For a fixed product $p = xy$, minimize $Ax + Bp/x$ on $p \le x \le 1$. For $p \le \rho$, its minimizer lies on $x = z, y = \rho z$. For $p > \rho$, it lies on $x = 1$. On that edge,

$$D(1, y) = -F_-(y)F_+(y),$$

where

$$F_- = -(\lambda+\alpha)^2 + \beta^2 + 4 + 4\beta(2y-1), \qquad F_+ = (\lambda-\alpha)^2 - \beta^2 - 4 + 4\beta(2y-1).$$

Since $\lambda \pm \alpha \geq 2 + |\beta|$, we have $F_- \leq 0 \leq F_+$. Thus it suffices to test the segment $(x, y) = (z, \rho z)$, $0 \leq z \leq 1$, for every choice of tilt signs.

Along this segment, put $R = 4z(1-z)$, $T = 4\rho z(1-\rho z)$, and $\Omega = 2 - 4\rho z^2$. Congruence by $\mathrm{diag}(\sqrt{RT}, \sqrt{R}, \sqrt{T}, 1)$ gives

$$P(z) = \begin{pmatrix} RT(\lambda-\Omega-\alpha-\beta) & -2zRT & -2\rho zRT & RT \\ -2zRT & R(\lambda+\Omega-\alpha+\beta) & RT & 2\rho zR \\ -2\rho zRT & RT & T(\lambda+\Omega+\alpha-\beta) & 2zT \\ RT & 2\rho zR & 2zT & \lambda-\Omega+\alpha+\beta \end{pmatrix}. \quad \text{(F10)}$$

This matrix has degree at most six. The congruence is invertible on $0 < z < 1$; the original slack at both endpoints follows by continuity from that interval. The matrix interval sum-of-squares theorem gives

$$P(z) = Y_3(z)^T U Y_3(z) + z(1-z) Y_2(z)^T V Y_2(z), \qquad U \succeq 0, \quad V \succeq 0, \quad \text{(F11)}$$

with $Y_j = (I_4, zI_4, \ldots, z^j I_4)^T$, and block orders $16, 12$. One can obtain this interval theorem directly from the real-line matrix theorem: apply it to $(1+t^2)^6 P(t^2/(1+t^2))$, split its factors into even and odd parts, and divide back. Even factors become polynomials of degree at most three in z; odd factors give $\sqrt{z(1-z)}$ times polynomials of degree at most two. This argument also includes singular matrices and endpoint zeros.

For fixed α, β, coefficient matching in Eq. (F11) is affine in λ, U, V. Include $\lambda \geq 2 + q$ explicitly. This yields an exact SDP of fixed size whenever $|\alpha| + |\beta| < 2$, including either zero-tilt axis and CHSH. For $|\alpha| + |\beta| \geq 2$, the local-bound certificates above apply. The ratio ρ still depends nonlinearly on the two independent tilts: this construction computes their support values, while the common-tilt construction above gives an affine membership lift.

F5 Thirty-two exact projections and their NPA intersections

Write $a_x = \langle A_x \rangle$, $b_y = \langle B_y \rangle$, and $E_{ij} = \langle A_i B_j \rangle$. For a sign pattern $\chi \in \{\pm 1\}^{2\times 2}$ satisfying $\prod_{ij} \chi_{ij} = -1$, and $x, y \in \{0, 1\}$, set

$$M_{\chi,xy} = a_x + \chi_{xy} b_y, \qquad S_\chi = \sum_{ij} \chi_{ij} E_{ij}.$$

The quantum image of $(M_{\chi,xy}, S_\chi)$ is exactly $\mathcal{W}$. Here is an explicit physical relabelling. In the canonical observables replace

$$A'_0 = A_x, \quad A'_1 = \varepsilon_1 A_{1-x}, \qquad B'_0 = \eta_0 B_y, \quad B'_1 = \eta_1 B_{1-y},$$

where

$$\eta_0 = \chi_{xy}, \qquad \eta_1 = \chi_{x,1-y}, \qquad \varepsilon_1 = \chi_{1-x,y}/\eta_0.$$

Then $a'_0 + b'_0 = M_{\chi,xy}$. The first three CHSH coefficients are the corresponding entries of χ, and the fourth is $-\varepsilon_1\eta_1 = \chi_{1-x,1-y}$ because the product of the four entries is -1. The relabelling is invertible and preserves quantum realizability, which proves equality of the images.

These are thirty-two distinct two-dimensional subspaces of the space of Bell functionals. The marginal part identifies x, y. Equality of the correlation directions would require $\chi' = \chi$ or $\chi' = -\chi$. The latter changes the relative sign of the two nonzero marginal coefficients, so the marginal directions are not proportional. Thus only the identical pair gives the same plane.

Let $\mathcal{Q}_L$ be a fixed standard NPA relaxation and define

$$\mathcal{R}_L = \mathcal{Q}_L \cap \bigcap_{\chi,x,y} \{p : \pi_{\chi,xy}(p) \in \mathcal{W}\}, \qquad \pi_{\chi,xy}(p) = (M_{\chi,xy}, S_\chi).$$

Every quantum correlation satisfies these constraints, and each projection of $\mathcal{R}_L$ lies in $\mathcal{W}$:

$$\mathcal{Q}_{222} \subseteq \mathcal{R}_L \subseteq \pi^{-1}_{\chi,xy}(\mathcal{W}).$$

For any real h, α, these two inclusions imply

$$\sup_{p\in\mathcal{R}_L} [hS_\chi(p) + \alpha M_{\chi,xy}(p)] = \sup_{p\in\mathcal{Q}_{222}} [hS_\chi(p) + \alpha M_{\chi,xy}(p)].$$

Part II gives directions on the canonical plane for which the level-L support is strictly larger. Hence $\mathcal{R}_L \subsetneq \mathcal{Q}_L$ for every finite L.

The rational data file `ALL32_EXACT_PROJECTION_CONSTRAINTS.json` implements each copy of Eqs. (F6)–(F7) with the two appropriate linear forms. The added constraints use 64 PSD blocks, alternating orders five and four, total order 288, 512 auxiliary variables and 96 affine equations. The generator enumerates physical relabellings and checks the distinct maps exactly. The inclusions above establish exactness on the stated planes; they do not establish quantum realizability of a full correlation whose projections satisfy them.

II. The complete finite-NPA obstruction

S1 Bell values, quantum models, and finite moment certificates

Alice and Bob choose $x, y \in \{0, 1\}$ and obtain $a, b \in \{\pm 1\}$. A valid correlation is a nonnegative, normalized, no-signaling table $p(a, b|x, y)$. Its four marginal expectations and four correlators determine it:

$$p(a, b|x, y) = \tfrac{1}{4}[1 + a\langle A_x\rangle + b\langle B_y\rangle + ab\langle A_x B_y\rangle]. \tag{S1}$$

Here the expectations are the averages of a, b, and ab under the corresponding marginals or joint distribution. For $0 \le \alpha < 1$, set

$$h_\alpha = A_0B_0 + A_0B_1 + A_1B_0 - A_1B_1 + \alpha(A_0 + B_0), \qquad \beta_\alpha(p) = \langle h_\alpha\rangle_p. \tag{S2}$$

The CHSH bound 2 and the two marginal bounds give the local maximum $\omega_{\mathrm{L}}(\alpha) = 2 + 2\alpha$, attained when all outputs are $+1$. In this part we abbreviate $\mathcal{Q}_{222}$ to $\mathcal{Q}$. Write $\omega_{\mathrm{Q}}(\alpha) = \max_{p\in\mathcal{Q}} \beta_\alpha(p)$ and $\Delta_\alpha = \omega_{\mathrm{Q}}(\alpha) - \omega_{\mathrm{L}}(\alpha)$, the quantum violation above the local bound.

The quantum set $\mathcal{Q}$ consists of $p(a, b|x, y) = \mathrm{Tr}[\rho(E_{a|x} \otimes F_{b|y})]$, with a density operator ρ and local binary positive-operator-valued measurements (POVMs), $\{E_{a|x}\}_a$ and $\{F_{b|y}\}_b$. A common local Naimark dilation preserves all probabilities and makes both settings projective [11]; the commuting-model version follows from Proposition 3.4(b) of Ref. [12]. The differences of the two outcome projectors are self-adjoint reflections, $A_x^2 = B_y^2 = I$, with projectors $(I + aA_x)/2$, $(I + bB_y)/2$.

Let $\mathcal{Q}_{\mathrm{fd}}$ denote the finite-dimensional tensor-product quantum set. Jordan's lemma decomposes each finite-dimensional local pair into invariant blocks of dimension at most two [13, 14, 15]. For block projections $P_{ij} = \Pi_i^A \otimes \Pi_j^B$, the weights $r_{ij} = \mathrm{Tr}(P_{ij}\rho)$ and states $P_{ij}\rho P_{ij}/r_{ij}$ give $p = \sum_{ij:r_{ij}>0} r_{ij}p_{ij}$. Conversely, correlated classical labels realize any finite mixture by local direct sums. If $\mathcal{K}_2$ denotes the two-qubit correlations with projective measurements, including the projectors 0 and I, then $\mathcal{Q}_{\mathrm{fd}} = \mathrm{conv}\,\mathcal{K}_2$. Carathéodory's theorem requires at most nine components in the eight-dimensional correlation space. The state and projector parameter sets are compact, so the mixture set is the compact continuous image of $\Delta_8 \times \mathcal{K}_2^9$, where Δ_8 is the simplex of nine nonnegative weights summing to one. Finite-dimensional approximation adds no tensor-product correlations. In the commuting model, denoted by $\mathcal{Q}_{\mathrm{c}}$, Alice's and Bob's operators act on one Hilbert space and commute with each other. The local two-projection algebra is nuclear [16, Sec. 4, Example 7(2)]; Proposition 3.4 and Remark 3.6 of Ref. [12] therefore identify the commuting set with the tensor-product closure. Thus, with complex Hilbert spaces throughout,

$$\mathcal{Q}_{\mathrm{fd}} = \mathcal{Q} = \overline{\mathcal{Q}_{\mathrm{fd}}} = \mathcal{Q}_{\mathrm{c}} = \mathrm{conv}\,\mathcal{K}_2. \tag{S3}$$

In particular, NPA converges to $\mathcal{Q}$ here [17, Theorem 8].

To define the finite NPA levels, we use formal observable products, called words. The group relations express that the observables square to the identity and that Alice's observables commute with Bob's. With e denoting the identity, the group is

$$G = \langle a_0, a_1, b_0, b_1 : a_x^2 = b_y^2 = e,\ a_x b_y = b_y a_x\rangle. \tag{S4}$$

Its complex group algebra consists of finite complex linear combinations of these words. Reduced words have Alice's alternating letters before Bob's alternating letters; the adjoint reverses a word and conjugates its coefficient. Let $\mathcal{W}_L$ contain the reduced words of length at most L, let $V_L = \mathrm{span}_{\mathbb{C}}\,\mathcal{W}_L$, and enumerate its words as $\boldsymbol{w}$. A candidate moment assignment associates a complex number $\ell(w)$ with each word through length $2L$, extends linearly, and is Hermitian when $\ell(w^*) = \overline{\ell(w)}$. It need not come from a physical realization. Its moment matrix and the resulting NPA bound are

$$\Gamma_L(\ell) = (\ell(u^*v))_{u,v\in\mathcal{W}_L}, \qquad \omega_{\mathrm{NPA}}^{(L)}(\alpha) = \max_{\ell(e)=1,\ \Gamma_L(\ell)\succeq 0} \ell(h_\alpha), \tag{S5}$$

where h_α now uses the abstract generators. For a measurement representation π and a unit state vector $|\psi\rangle$, physical moments $\ell(w) = \langle\psi|\pi(w)|\psi\rangle$ make Γ_L a Gram matrix; therefore $\omega_{\mathrm{NPA}}^{(L)} \geq \omega_{\mathrm{Q}}$. We call $\omega_{\mathrm{NPA}}^{(L)} - \omega_{\mathrm{Q}}$ the NPA approximation error. Level 1 is an observable moment bound and need not give nonnegative probabilities. We extract correlations only at levels $d \geq 2$ below.

An identity $\lambda e - h_\alpha = \sum_j p_j^* p_j$ in the group algebra, with $p_j \in V_L$, is an upper-bound certificate: evaluation on any measurements gives $\lambda I - \pi(h_\alpha) \succeq 0$. Its Gram matrix is the variable in the dual problem below.

Proposition S1 (Attainment and Gram trace bound)**.** *For every finite $L \geq 1$ and $0 \leq \alpha \leq 1$, the primal maximum in Eq.* (S5) *and the dual minimum below are attained:*

$$\omega_{\mathrm{NPA}}^{(L)}(\alpha) = \min\{\lambda \in \mathbb{R} : \lambda e - h_\alpha = \boldsymbol{w}^* Q \boldsymbol{w},\ Q \succeq 0\} \tag{S6}$$

The Gram matrix may be real symmetric. Every such certificate satisfies the following bounds, where matrix norms are operator norms:

$$\operatorname{Tr} Q = \lambda, \qquad \|Q\| \leq \lambda, \qquad \|Q^{1/2}\| \leq \sqrt{\lambda}. \tag{S7}$$

Proof. The canonical trace $\tau(\sum_g c_g g) = c_e$ obeys $\tau(u^* v) = \delta_{uv}$, so $\Gamma_L(\tau) = I \succ 0$. For feasible ℓ, the two-by-two principal minors give $|\ell(u^* v)|^2 \leq \ell(u^* u)\ell(v^* v) = 1$. Every word through length $2L$ is $u^* v$ with $u, v \in \mathcal{W}_L$ (split the word into two pieces of length at most L and invert the first). Thus the feasible set is closed and bounded in a finite-dimensional space, so the maximum is attained. Strict feasibility and finiteness give dual attainment and equality by Slater's theorem [19]. The dual identity is precisely Eq. (S6): writing $Q = C^* C$ gives a sum $\sum_j p_j^* p_j$ with $p_j \in V_L$, and conversely every such sum gives a positive Gram matrix. The multiplication table and h_α have real coefficients, so averaging Q with $\overline{Q}$ preserves the identity and positivity. Applying τ gives $\lambda = \sum_{ij} Q_{ij} \tau(w_i^* w_j) = \operatorname{Tr} Q$, since $\tau(h_\alpha) = 0$; positivity gives the norm bounds. The moment bounds also imply $\omega_{\mathrm{NPA}}^{(L)}(\alpha) \leq 4 + 2\alpha \leq 6$. □

For comparison, $\mathcal{W}_1 = \{e, a_0, a_1, b_0, b_1\}$ already recovers the CHSH maximum $2\sqrt{2}$ [6, 7]. The list $\mathcal{W}_{1+AB} = \mathcal{W}_1 \cup \{a_x b_y\}_{x,y}$ gives tight certificates for one-sided tilted CHSH, $h_0 + \gamma a_0$ for $0 \leq \gamma < 2$, and self-tests its maximizers [8, 9, 10]. Thus level $1 + AB$ is exact when either marginal tilt vanishes [1].

S2 The cubic quantum violation

On a nontrivial Jordan block local basis choices give $A_0 = B_0 = Z$, $A_1 = cZ + rX$, $B_1 = dZ + sX$, where $c, d \in [-1, 1]$, $r = \sqrt{1 - c^2}$, and $s = \sqrt{1 - d^2}$. The endpoints include one-dimensional blocks: the assignment $(A_0, A_1) = (\epsilon_0, \epsilon_1)$ occurs on the Z eigenspace ϵ_0 at $c = \epsilon_0 \epsilon_1$, and similarly for Bob. Put $\alpha = 1 - T$, $\lambda_0(T) = 4 - 2T$, $u = 1 - c$, $v = 1 - d$, and $\Omega = 2 - uv$. The evaluated Bell operator is

$$H_T(c, d) = \begin{pmatrix} \Omega + 2(1 - T) & us & vr & -rs \\ us & -\Omega & -rs & -vr \\ vr & -rs & -\Omega & -us \\ -rs & -vr & -us & \Omega - 2(1 - T) \end{pmatrix}. \tag{S8}$$

Equation (S3) gives $\omega_{\mathrm{Q}}(1 - T) = \max_{c,d \in [-1,1]} \lambda_{\max} H_T(c, d)$, attained on two qubits. For later use, multiplication by a real vector $\eta = (1, b_1, b_2, b_3)^T$ yields the exact identity, also valid for signed r, s with $r^2 = 1 - c^2$, $s^2 = 1 - d^2$,

$$\begin{aligned} \mathcal{E}_T(\eta) &:= \langle \eta, (H_T - \lambda_0(T) I)\eta \rangle \\ &= -uv + 2usb_1 + 2vrb_2 - 2rsb_3 + (-6 + 2T + uv)(b_1^2 + b_2^2) \\ &\quad + (-4 + 4T - uv) b_3^2 - 2rsb_1 b_2 - 2vrb_1 b_3 - 2usb_2 b_3. \end{aligned} \tag{S9}$$

The normalized vector's Bell excess is $\mathcal{E}_T(\eta)/\|\eta\|^2$.

Theorem 1 and Eqs. (18)–(19) of Ref. [1] identify $\omega_{\mathrm{Q}}(\alpha)$ as the largest real root in λ of

$$F(\lambda,\alpha) = \lambda^4 + (4-\alpha^2)\lambda^3 + (\tfrac{11}{4}\alpha^4 - 12\alpha^2 - 4)\lambda^2 + (2\alpha^6 - \alpha^4 - 20\alpha^2 - 32)\lambda + 5\alpha^6 - 21\alpha^4 + 16\alpha^2 - 32, \tag{S10}$$

with common optimal cosine

$$c_*(\alpha) = \tfrac{1}{8}[3\alpha^2 - 4 + \sqrt{16 + 9\alpha^4 + 8\alpha^2(2\omega_{\mathrm{Q}}(\alpha) - 1)}]. \tag{S11}$$

At $\alpha = 1$, $F(\lambda, 1) = (\lambda - 4)(4\lambda^3 + 28\lambda^2 + 59\lambda + 32)/4$. The cubic is positive for $\lambda \geq 4$. The local lower bound and $\omega_{\mathrm{Q}}(\alpha) \leq 4 + 2\alpha$ force every endpoint limit to be the root 4. Changing the tilt by T changes any Bell value by at most $2T$, so $\omega_{\mathrm{Q}}(1-T) = 4 + O(T)$. Since

$$F(2+2\alpha, \alpha) = 4(\alpha-1)^3(\alpha+2)^4, \qquad \partial_\lambda F(4,1) = 243,$$

the mean-value theorem at fixed $\alpha = 1 - T$ gives

$$\Delta_{1-T} = \frac{4T^3(3-T)^4}{\partial_\lambda F(\widetilde{\lambda}_T, 1-T)} = \frac{324}{243}T^3 + O(T^4) = \frac{4}{3}T^3 + O(T^4), \tag{S12}$$

where $\widetilde{\lambda}_T$ lies between the local and quantum values. The nonzero derivative also gives a unique analytic root near 4. Writing this root as $4 - 2T + \frac{4}{3}T^3 + c_4T^4 + c_5T^5 + O(T^6)$ and substituting into $F = 0$ gives $27(9c_4 + 4)T^4 + (243c_5 - 243c_4 - 120)T^5 + O(T^6) = 0$. Consequently

$$\omega_{\mathrm{Q}}(1-T) = 4 - 2T + \tfrac{4}{3}T^3 - \tfrac{4}{9}T^4 + \tfrac{4}{81}T^5 + O(T^6), \tag{S13}$$

$$c_*(1-T) = 1 - 2T + \tfrac{8}{9}T^2 + \tfrac{20}{81}T^3 + O(T^4). \tag{S14}$$

Analyticity gives constants $C, \delta > 0$ bounding each remainder by CT^k at its displayed order for $0 \leq T < \delta$. In particular the quartic correction to the quantum value divided by T^4 tends to $-4/9$. Since $1 - \cos\theta = \theta^2/2 + O(\theta^4)$, the optimal angles obey $\theta_* \sim 2\sqrt{T}$; the cited self-testing theorem gives this scale for every maximizer in the canonical form.

This endpoint has a detector-loss interpretation [4, 1]. Let each detector fire independently with common probability η, independently also of settings and ideal outcomes, and assign $+1$ on no-detection. Then

$$\langle A_x B_y\rangle_\eta = \eta^2\langle A_x B_y\rangle + \eta(1-\eta)(\langle A_x\rangle + \langle B_y\rangle) + (1-\eta)^2,$$
$$\mathrm{CHSH}_\eta(p) = \eta^2\beta_\alpha(p) + 2(1-\eta)^2, \qquad \alpha = 2(1-\eta)/\eta.$$

The setting-1 marginals cancel in CHSH. The local bound becomes 2, and both the quantum violation and the NPA error are multiplied by η^2, preserving their ratio. Here $\alpha \uparrow 1$ corresponds to $\eta \downarrow 2/3$ [5]; since $T = (3\eta - 2)/\eta = (9/2)(\eta - 2/3) + O((\eta - 2/3)^2)$,

$$\eta^2\Delta_\alpha = 54(\eta - 2/3)^3 + O((\eta - 2/3)^4). \tag{S15}$$

S3 A two-parameter trial state and the Motzkin polynomial

The quantum violation is of order T^3, and the optimal measurement angles are of order $\sqrt{T}$. Set $T = t^2$ so that these scales become t^6 and t, and vary the two angles independently:

$$A_0 = B_0 = Z, \quad A_1 = \cos(tx)Z + \sin(tx)X, \quad B_1 = \cos(ty)Z + \sin(ty)X, \qquad x, y \in \mathbb{R}. \tag{S16}$$

Write $H(t; x, y)$ for the evaluation of h_{1-t^2} on these observables. A certificate must hold for every (x, y), although the canonical rescaled optimal angles approach $(x, y) = (2, 2)$. In what follows

$O_K(t^k)$ bounds the scalar, Euclidean, or operator norm uniformly on each compact $K \subset \mathbb{R}^2$ by $C_K|t|^k$ for $0 < |t| < \delta_K$.

At $t = 0$, $H = \mathrm{diag}(4, -2, -2, 0)$; the gaps from $|00\rangle$ are $6, 6, 4$. Its couplings to the other basis vectors are $x^2yt^3/2 + O_K(t^5)$, $xy^2t^3/2 + O_K(t^5)$, and $-xyt^2 + O_K(t^4)$. Take $\zeta_t = (1, At^3, Bt^3, Ct^2)^T$ and $\Lambda_t = 4 - 2t^2 + 4t^6/3$. We choose A, B, C to make the expectation of $\Lambda_t I - H(t)$ vanish below order t^6. Equation (S9) gives the leading gap $(4C + xy)^2t^4/4$. Hence choose $C = -xy/4$. The next coefficient is

$$\langle \zeta_t, (\Lambda_t I - H(t))\zeta_t \rangle = t^6 \Big[6(A - x^2y/12)^2 + 6(B - xy^2/12)^2 + \tfrac{4}{3} + \frac{x^4y^2 + x^2y^4}{48} - \frac{x^2y^2}{4} \Big] + O_K(t^8). \tag{S17}$$

Indeed, before completing the two squares their contribution is $6A^2 - Ax^2y + 6B^2 - Bxy^2$, and the coefficient of $x^4y^2 + x^2y^4$ is $1/16$; completing the squares subtracts $1/24$, leaving $1/48$. Choose $A = x^2y/12$ and $B = xy^2/12$. Define the resulting polynomial vector ξ_t and the Motzkin polynomial $M(X, Y)$ by

$$\xi_t = (1, t^3x^2y/12, t^3xy^2/12, -t^2xy/4)^T, \qquad M(X, Y) = X^4Y^2 + X^2Y^4 - 3X^2Y^2 + 1 \tag{S18}$$

Then ξ_t satisfies

$$\langle \xi_t, [\omega_{\mathrm{Q}}(1 - t^2)I - H(t; x, y)]\xi_t \rangle = \tfrac{4}{3}t^6 M(x/2, y/2) + O_K(t^8). \tag{S19}$$

To check the remainder, sine and cosine have uniform Taylor remainders on K. Conjugation by $U = Z \otimes Z$ gives $H(-t) = UH(t)U$ and $\xi_{-t} = U\xi_t$, so the expectation is even. Replacing Λ_t by the exact quantum value changes it by $[-4t^8/9 + 4t^{10}/81 + O(t^{12})]\|\xi_t\|^2 = O_K(t^8)$, since

$$\|\xi_t\|^2 = 1 + x^2y^2t^4/16 + (x^4y^2 + x^2y^4)t^6/144. \tag{S20}$$

Equivalently, ξ_t is the top eigenvector, with first component one, through order three: its leading component equations are $6A = x^2y/2$, $6B = xy^2/2$, $4C = -xy$. The simple eigenvalue at $t = 0$ and its fixed spectral gap give an $O_K(t^4)$ vector remainder.

The normalized state $\psi_t = \xi_t/\|\xi_t\|$ has Bell value $b_t = \langle \psi_t, H(t)\psi_t \rangle$, and

$$\frac{\omega_{\mathrm{Q}}(1 - t^2) - b_t(x, y)}{\Delta_{1-t^2}} \longrightarrow M(x/2, y/2) \tag{S21}$$

locally uniformly. On either axis $\xi_t = |00\rangle$ is an eigenvector at the local bound, so this ratio is exactly 1. At $(x, y) = (2, 2)$ it tends to zero. The gap in this formula is a physical strategy's gap from the quantum bound; the NPA error will add a separate nonnegative term.

Lemma S2 (Shifted Motzkin polynomial)**.** *M is nonnegative, but $M + c$ is not a sum of real polynomial squares for any $c \geq 0$.*

Proof. Arithmetic–geometric mean gives $X^4Y^2 + X^2Y^4 + 1 \geq 3X^2Y^2$. If $M + c = \sum_j q_j^2$, the highest homogeneous terms cannot cancel, so $\deg q_j \leq 3$. On either axis the sum is the constant $1 + c$; each restricted polynomial is bounded and therefore constant. Thus $q_j = a_j + b_jXY + c_jX^2Y + d_jXY^2$. The X^2Y^2 coefficient of $\sum_j q_j^2$ is $\sum_j b_j^2 \geq 0$, contradicting -3. □

An alternative proof uses the Newton polytope $\mathrm{Newt}(p)$, the convex hull of the exponent vectors of monomials with nonzero coefficients in p. The usual half-Newton-polytope property gives the same monomial restriction: if $p = \sum_j q_j^2$, then $\mathrm{Newt}(q_j) \subseteq \frac{1}{2}\mathrm{Newt}(p)$ [23, Theorem 1]. The homogeneous statement gives the polynomial statement by homogenization and then setting the extra variable to one. Here $\mathrm{Newt}(M + c) = \mathrm{conv}\{(0, 0), (4, 2), (2, 4)\}$, and its half has integer points $(0, 0), (1, 1), (2, 1), (1, 2)$, exactly the four monomials above. Independent angles are essential: the restriction $M(X, X) = (X^2 - 1)^2 + 2[X(X^2 - 1)]^2$, even after adding $c \geq 0$, is a sum of squares.

S4 Polynomial limits of fixed certificates and divergence

An evaluated certificate gives squared norms of analytic functions. To obtain polynomial square factors in its limit, we first control the Taylor remainder and then bound coefficients in one fixed polynomial space. For a fixed list $\mathcal{W}_L = (w_1, \dots, w_N)$, let

$$z_t(x,y) = (\pi_{t,x,y}(w_i)\xi_t(x,y))_{i=1}^N \in \mathbb{C}^{4N}, \tag{S22}$$

where π evaluates Eq. (S16).

Lemma S3 (Fixed-level sum-of-squares limit)**.** *Let $t_n \downarrow 0$, $Q_n \succeq 0$ be $N \times N$ matrices with $\sup_n \operatorname{Tr} Q_n < \infty$, and $m \geq 0$ be an integer. If*

$$t_n^{-2m} \|(Q_n^{1/2} \otimes I_4) z_{t_n}(x,y)\|^2 \longrightarrow P(x,y) \tag{S23}$$

locally uniformly for a real polynomial P, then P is a sum of squares of real polynomials.

Proof. A sine or cosine coefficient of order k in t is a polynomial in x, y of total degree at most k; the entries of ξ_t have the same property. Multiplication adds these orders. Since the word list is fixed and finite, Taylor's theorem gives, on every compact K,

$$z_t = \sum_{r=0}^{m} t^r z_r + R_{m+1}, \qquad \deg z_r \leq r, \qquad \sup_K \|R_{m+1}\| \leq C_K |t|^{m+1}. \tag{S24}$$

Put $C_n = Q_n^{1/2}$, keeping all N rows even if its rank changes. The trace bound gives $\|C_n\| \leq \sqrt{\sup_n \operatorname{Tr} Q_n}$. Define the polynomial vector p_n of degree at most m and the analytic vector v_n by

$$p_n = t_n^{-m}(C_n \otimes I_4) \sum_{r=0}^{m} t_n^r z_r, \qquad v_n = t_n^{-m}(C_n \otimes I_4) z_{t_n}, \qquad \sup_K \|p_n - v_n\| = O_K(t_n), \tag{S25}$$

The vectors p_n are uniformly bounded on $K_0 = [-1,1]^2$, since $\|v_n\|^2 \to P$ uniformly there. To bound coefficients despite the negative powers of t_n, choose $m+1$ fixed distinct nodes $s_i \in [-1,1]$. Their fixed Lagrange polynomials L_i give, for each coordinate p of p_n,

$$p(x,y) = \sum_{i,j=0}^{m} p(s_i, s_j) L_i(x) L_j(y).$$

The bounded grid values therefore bound every coefficient of p_n. One subsequence makes all finitely many complex coefficients converge, giving a polynomial vector q and locally uniform convergence $p_n \to q$. On K_0, $|\|p_n\|^2 - \|v_n\|^2| \leq (\|p_n\| + \|v_n\|)\|p_n - v_n\| \to 0$, so $P = \|q\|^2$ there. This is a polynomial identity on an open set and thus on $\mathbb{R}^2$. Splitting each coordinate of q into real and imaginary parts expresses P as real polynomial squares. □

Theorem S4. *For every fixed finite $L \geq 1$,*

$$\lim_{\alpha \uparrow 1} \frac{\omega_{\mathrm{NPA}}^{(L)}(\alpha) - \omega_{\mathrm{Q}}(\alpha)}{(1-\alpha)^3} = +\infty, \qquad \lim_{\alpha \uparrow 1} \frac{\omega_{\mathrm{NPA}}^{(L)}(\alpha) - \omega_{\mathrm{Q}}(\alpha)}{\Delta_\alpha} = +\infty. \tag{S26}$$

Proof. If the first divergence failed, nonnegativity of the error would give $t_n \downarrow 0$ with $\omega_{\mathrm{NPA}}^{(L)}(1-t_n^2) = \omega_{\mathrm{Q}}(1-t_n^2) + r_n t_n^6$, where, after a subsequence, $r_n \to r \geq 0$. Proposition S1 gives exact Gram certificates

$$\omega_{\mathrm{NPA}}^{(L)}(1-t_n^2)e - h_{1-t_n^2} = \boldsymbol{w}^* Q_n \boldsymbol{w}, \qquad \operatorname{Tr} Q_n = \omega_{\mathrm{NPA}}^{(L)}(1-t_n^2) \leq 6.$$

Evaluate on the observables and ξ_{t_n}, then split the expectation into the quantum strategy gap and $r_n t_n^6 \|\xi_{t_n}\|^2$. Equations (S19) and (S20) give, locally uniformly,

$$t_n^{-6} \|(Q_n^{1/2} \otimes I_4) z_{t_n}\|^2 \longrightarrow \tfrac{4}{3} M(x/2, y/2) + r. \tag{S27}$$

Lemma S3 with $m = 3$ makes this polynomial a sum of real squares. Substituting $(x,y) = (2X, 2Y)$ and multiplying by $3/4$ would express $M(X,Y) + 3r/4$ as such a sum, contrary to Lemma S2. The second limit follows from $\Delta_\alpha/(1-\alpha)^3 \to 4/3$. □

The absolute NPA error remains bounded. The divergence concerns its ratio to a quantum violation that vanishes cubically. For each fixed L it also gives a full interval $1-\delta_L<\alpha<1$ with a strict gap; δ_L may depend on L. No assertion that one fixed $\alpha<1$ is missed at every finite level is needed.

S5 Valid correlations, arbitrary finite word lists, and a local limit

Fix any finite list $\mathcal{S}=(S_1,\dots,S_k)$ of words in the measurement projectors. Define $\mathcal{N}_{\mathcal{S}}$ to contain the valid no-signaling correlations for which a normalized Hermitian functional ℓ satisfies

$$\Gamma_{\mathcal{S}}=(\ell(S_i^*S_j))_{ij}\succeq 0,\qquad p(a,b|x,y)=\ell(E_{a|x}F_{b|y}), \tag{S28}$$

together with the identities generated by $E_{a|x}^*=E_{a|x}$, $E_{a|x}E_{a'|x}=\delta_{aa'}E_{a|x}$, $\sum_a E_{a|x}=I$, the analogous Bob relations, and $[E_{a|x},F_{b|y}]=0$. The functional is defined on all finitely many words needed for these constraints and the observed probabilities; $\ell(W^*)=\overline{\ell(W)}$. If a moment is absent from the positive matrix it is free apart from the stated identities. This definition includes standard finite NPA lists; additional localizing matrices, state- or operator-optimality constraints, or problem-specific inequalities define other relaxations.

Lemma S5. *Every level-d feasible functional with $d\geq 2$ defines a valid correlation through $E_{a|x}=(e+aa_x)/2$, $F_{b|y}=(e+bb_y)/2$ and $p(a,b|x,y)=\ell(E_{a|x}F_{b|y})$. Writing $\mathcal{N}_d$ for these extracted correlations, every fixed finite list $\mathcal{S}$ admits a finite $d\geq 2$ such that $\mathcal{N}_d\subseteq\mathcal{N}_{\mathcal{S}}$.*

Proof. $P=E_{a|x}F_{b|y}$ is a self-adjoint projector in $V_2\subseteq V_d$, so $p=\ell(P)=\ell(P^*P)\geq 0$. Summing the projectors gives $\sum_{ab}p=\ell(e)=1$ and $\sum_b p=\ell(E_{a|x})$, independently of y; Bob's marginal is analogous. This proves validity and preserves the moments $\langle A_x\rangle_p=\ell(a_x)$, $\langle B_y\rangle_p=\ell(b_y)$, and $\langle A_xB_y\rangle_p=\ell(a_xb_y)$.

Expand each projector word using these affine formulas. A word of D projector factors expands into observable words of length at most D; reduction only shortens them. Choose $d\geq 2$ at least the largest such length and enlarge it for any other finitely many required moments. Then $\boldsymbol{S}=T\boldsymbol{w}_d$ for a real matrix T, and

$$\Gamma_{\mathcal{S}}=T\Gamma_d T^*\succeq 0. \tag{S29}$$

The group relations imply all projector identities. Any remaining unconstrained moments admit a Hermitian linear extension on the reduced-word basis, assigning conjugate values on inverse pairs. No further positive matrix involving those moments is required. Hence the extracted correlation belongs to $\mathcal{N}_{\mathcal{S}}$. □

Theorem S6. *For every fixed finite projector-word list $\mathcal{S}$ as above, $\mathcal{Q}\subsetneq\mathcal{N}_{\mathcal{S}}$. Moreover, every neighborhood of $p_{\mathrm{loc}}(a,b|x,y)=\delta_{a,1}\delta_{b,1}$ contains a correlation in $\mathcal{N}_{\mathcal{S}}\setminus\mathcal{Q}$.*

Proof. Quantum moments satisfy all the constraints. Choose $d\geq 2$ from the lemma and a primal optimizer at each α, extracting $p_{d,\alpha}\in\mathcal{N}_d\subseteq\mathcal{N}_{\mathcal{S}}$ with $\beta_\alpha(p_{d,\alpha})=\omega_{\mathrm{NPA}}^{(d)}(\alpha)$. For α sufficiently close to one, Theorem S4 puts this value strictly above $\omega_{\mathrm{Q}}(\alpha)$, proving strict containment. Set

$$e_{d,\alpha}=\omega_{\mathrm{NPA}}^{(d)}(\alpha)-\omega_{\mathrm{Q}}(\alpha),\qquad \lambda_\alpha=\frac{2\Delta_\alpha}{\Delta_\alpha+e_{d,\alpha}},\qquad q_\alpha=(1-\lambda_\alpha)p_{\mathrm{loc}}+\lambda_\alpha p_{d,\alpha}. \tag{S30}$$

Since $e_{d,\alpha}/\Delta_\alpha\to\infty$, the weight is in $(0,1)$ near one and tends to zero. Convexity puts q_α in $\mathcal{N}_{\mathcal{S}}$, whereas $\beta_\alpha(q_\alpha)=\omega_{\mathrm{L}}(\alpha)+2\Delta_\alpha=\omega_{\mathrm{Q}}(\alpha)+\Delta_\alpha>\omega_{\mathrm{Q}}(\alpha)$. Thus $q_\alpha\notin\mathcal{Q}$. Every probability coordinate differs from p_{loc} by at most λ_α, proving convergence to the deterministic correlation. □

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